

SELF INTRODUCTION

Good morning, my name is Gayatri Bhoyar. I am an aspiring Data Analyst with a strong foundation in Python, SQL, Advanced Excel, and Power BI, and a genuine passion for working with data.

I have completed my Master's in Commerce and recently trained as a Data Analyst, where I gained hands-on experience in data cleaning, exploratory data analysis, and creating interactive dashboards that turn complex data into clear, actionable insights.

Before moving into data analytics, I worked for 2.5 years as a Quality Analyst at Tata Consultancy Services, where I developed strong analytical thinking, attention to detail, and a process-driven mindset. This professional experience helps me understand business requirements and analyse data with a practical, results-oriented approach.

I am highly motivated, quick to learn, and excited to begin my career as a Data Analyst where I can contribute my skills, grow continuously, and create impact through data.

Why should we hire you?

You should hire me because I bring a strong foundation in data analysis along with genuine eagerness to learn and grow. As a fresher, I may not have years of experience, but I make up for it with solid skills in data cleaning, analysis, and visualization, and a mindset that focuses on accuracy and problem-solving. I have worked on hands on projects where I turned raw data into meaningful insights, and I enjoy asking the right questions behind the numbers. I'm detail-oriented, comfortable working with tools like Excel, SQL, Python, and Power BI and I'm always improving my skills through practice and learning.

OR I am a fresher in the Data Analyst role, I have strong skills in tools like Excel, SQL, Python, and Power BI. During my training and projects, I worked on data cleaning, analysis, and creating dashboards to find useful insights from data. I am a quick learner, hardworking, and eager to apply my data analysis skills to help the company make better data-driven decisions.

What is your greatest strength?

My greatest strength is my analytical thinking and attention to detail. I enjoy working with data, identifying patterns, and finding insights that can help in decision-making. My experience as a Quality Analyst also helped me develop accuracy and problem-solving skills. I always ensure the data is clean and reliable before analysis, which helps in producing meaningful results.

What is your greatest weakness?

One of my weaknesses is that sometimes I spend extra time checking my work because I want the data and analysis to be completely accurate. However, I am improving this by managing my time better and setting clear deadlines for tasks. This helps me maintain both accuracy and efficiency.

Where do you see yourself in 5 years?

In the next five years, I see myself growing into a highly skilled Data Analyst who can handle complex datasets and provide valuable insights for business decisions. I want to strengthen my expertise in tools like SQL, Python, and Power BI and work on impactful data-driven projects. I also aim to take on more responsibilities and contribute to the organization's growth.

Why do you leave your last job?

In the Procure to Pay process, my role as a Quality Check Analyst was to audit purchase and payment transactions, verify invoices and purchase orders, identify errors, and ensure the process followed company standards. I gained valuable experience as a Quality Check Analyst in the Procure to Pay process, but I developed a strong interest in data analysis. I upskilled myself in tools like Excel, SQL, Python, and Power BI, and now I am looking for an opportunity to start my career as a Data Analyst.

General Interview Questions

1. What is Data Analysis?

Data analysis is basically a process of analysing, modelling, and interpreting data from multiple sources to identify patterns, trends, and valuable insights. It helps organizations understand their current market position, identify problems, and make informed decisions for improvement.

2. What does a Data Analyst do?

A data analyst collects and studies data to help companies make informed decisions. They gather data from different sources, filter and organize it, and search for patterns or trends. They present and forecast their findings clearly so teams can improve performance or solve business issues.

3. Why do You Want to Pursue a Career in Data Analytics?

I want to pursue a career in data analytics because I genuinely enjoy working with data and solving problems. I like looking at data, finding

patterns, and understanding what it actually means. When those insights help a business make better decisions, it feels very satisfying. I also like that this field keeps changing, so I get to learn and improve my skills continuously.

4. What is Structured and Unstructured Data?

Structured data is well-organized data stored in tables, spreadsheets, or databases, which makes it easy to search and analyse, such as customer details or sales records. Unstructured data does not follow a fixed format and includes things like emails, images, videos, and social media content, which are harder to analyse using traditional tools.

5. Why is Excel Important for Data Analysts?

Excel is essential for data analysts because it helps organize and clean large datasets, perform calculations, and spot trends quickly using various tools, functions, and formulas. It is cost-effective and widely available, which makes it easy to use for preliminary analysis before deeper investigations with more complex tools.

6. What are Pivot Tables?

Pivot tables are used to quickly summarize and analyse large amounts of data in tools like Excel or Google Sheets. They help organize data by grouping it into rows and columns so patterns and totals are easy to see. I can use pivot tables to calculate values like totals, averages, or counts for different categories. Since they work with drag-and-drop, they make reporting fast and easy without complex formulas.

7. What is SQL Used for?

SQL (Structured Query Language) is used to access, manage, and manipulate data stored in relational databases. Data analysts use SQL to retrieve specific information using SELECT statements, combine data from multiple tables using JOINS, filter records with WHERE

clauses, and perform calculations and groupings with functions like SUM, COUNT, and GROUP BY.

8. What is Data Cleaning?

Data cleaning is the process of identifying and removing misleading or inaccurate records from the datasets. The primary objective of data cleaning is to improve data quality so it can be used for analysis and predictive modelling. It is the next process after data collection and loading.

9. What is Data Validation, and how do You Check Data Accuracy?

Data validation is the process of checking whether data is accurate, complete, and usable before analysis. Analysts verify data types, ranges, and formats to ensure that values conform to predefined rules. Accuracy is checked by comparing data against source records and identifying inconsistencies, which helps maintain data integrity and prevents errors.

10. What is Data Visualization?

Data visualization is the practice of presenting data using charts, graphs, or other visual representations to make information easier to understand. It helps identify patterns, trends, comparisons, and outliers quickly. Effective visualization relies on clean data, choosing the correct chart for the message, and keeping the design uncluttered.

11. Which Chart is Best for Trend Analysis?

A line chart is best for trend analysis because it shows how values change over time. It makes it easy to spot patterns such as upward or downward movement, seasonality, or sudden changes. This helps analysts and stakeholders quickly understand how performance is evolving.

12. What Skills do You Think are Most Important for a Data Analyst?

Important skills for a data analyst include strong problem-solving, attention to detail, and the ability to identify patterns in data. Communication skills are essential for clearly explaining insights and supporting decision-making. Proficiency in data cleaning and data visualization, as well as basic programming/scripting, is also important for solving real-world problems.

13. What have You Learned Recently to Improve Your Data Analysis Skills?

Recently, I completed an online course on exploratory data analysis, which helped me approach data more systematically before analysis. I attended a webinar on building dashboards, learning to choose visuals based on audience and goals. I practiced applying these concepts to sample datasets to reinforce what I had learned.

14. How do You Ensure Data Accuracy in Your Analysis?

I ensure accuracy by validating data at multiple stages of the analysis process. I begin by checking for missing values, then cross-verify key metrics against source systems or existing reports to confirm their reliability. Before sharing results, I review assumptions, calculations, and outputs to ensure the insights align with the business context and are accurate.

16. What is Data Wrangling?

Data wrangling is closely related to preprocessing and is also known as data munging. It involves cleaning, transforming, and organizing raw or unstructured information into a format suitable for analysis. The main objective is to improve the quality and structure so it can be effectively used for reporting, analysis, or model building.

17. How do You Handle Large Datasets Efficiently?

When working with large datasets, I focus on filtering data early to include only relevant columns and time periods. I summarize data before analysis, avoid unnecessary calculations, and break the work

into smaller, manageable steps. This approach improves performance and keeps the evaluation focused on business goals.

18. How do You Turn a Business Problem into a Data Problem?

I begin by understanding the business objective and the decision to support. I then break the problem into specific questions that can be addressed through analysis, such as identifying trends, gaps, or changes in performance over time. This helps define the right metrics, data sources, and scope before the analysis begins.

19. How do You Decide which Data is Important for Analysis?

I focus on data that is directly relevant to the problem being analysed. I consider factors such as relevance, time period, data accuracy, and potential impact on decision-making. Any data that does not add value to the objective is excluded to keep the analysis efficient and clear.

24. What is the Difference between WHERE and HAVING?

The main difference between WHERE and HAVING is that WHERE filters individual rows before grouping or aggregation, while HAVING filters grouped results after aggregation. The WHERE clause is used for row-level conditions and cannot be used with aggregate functions directly. In practice, WHERE is used to limit the data early, and HAVING is used to refine aggregated results.

26. When do You Choose Python Instead of Excel?

Python is ideal for automation and building consistent, reusable workflows; tasks that would require manual effort in Excel can be scripted and scheduled in Python. I also prefer Python for complex statistical analysis, machine learning, and advanced data transformations beyond Excel's built-in functions. Python libraries like pandas, NumPy, and scikit-learn provide powerful capabilities for data manipulation and predictive modelling.

27. What tools are Commonly Used for Data Analysis?

Commonly used data analysis tools include Excel and SQL for data cleaning, exploration, and querying structured data. Programming languages such as Python are used for deeper analysis, automation, and statistical modelling. Visualization tools like Tableau or Power BI help present insights in a clear and visual format for stakeholders. The choice of tools usually depends on the data size, complexity, and project business requirements.

29. What Steps do You Follow to Analyse a Dataset?

I begin by understanding the business problem and the questions that need to be answered. I then collect relevant data and clean it to handle missing values, errors, and inconsistencies. Next, I perform exploratory data analysis to identify patterns, trends, and key drivers. Finally, I use clear visuals and explanations to present insights so stakeholders can easily understand the findings and take action.

30. What is the Importance of Exploratory Data Analysis (EDA)?

Exploratory data analysis is important because it helps understand the structure and behaviour of the data early on. It allows analysts to identify trends, relationships, and anomalies that may not be obvious at first glance. EDA also helps detect data quality issues such as missing values or outliers, before deeper analysis begins. This early understanding guides the choice of analytical methods and improves the accuracy and reliability of the final results.

31. What is the Difference between Descriptive and Predictive Analysis?

Descriptive analysis focuses on understanding past events by summarizing historical data. It explains what happened, identifying patterns, trends, and relationships through aggregation, visualization, and basic statistics. Predictive analysis builds on this historical data to forecast future outcomes. It relies on methods like regression, time series analysis, and machine learning. It's used for tasks like product

recommendations and predicting patient risk, playing a key role in data-driven decision-making.

32. What are Univariate, Bivariate, and Multivariate Analyses?

Univariate analysis focuses on a single variable and is used to describe its distribution, such as average sales or age distribution. Bivariate analysis examines the relationship between two variables, for example, how price changes affect sales. Multivariate analysis examines multiple variables together to identify patterns and dependencies, such as how price, marketing spends, and seasonality jointly influence revenue.

33. Why Should We Hire You for this Data Analyst Role?

I merge analytical skills and practical business understanding into the data analyst role. Along with working confidently with data and tools, I focus on clear communication and collaboration with different teams. I am careful about data accuracy and take ownership of my work. This approach ensures that analysis is practical, trusted, and supports real business decisions.

34. How do You Prioritize Multiple Data Requests?

When handling multiple data requests, I first assess the urgency and potential business impact of each. I also consider dependencies, such as whether one analysis is required before another can move forward. Based on this, I establish a clear execution order and set realistic timelines. Throughout the process, I keep stakeholders informed about progress and any changes to delivery expectations.

What are the different types of data?

Data can be categorized as: Quantitative (numerical) vs Qualitative (categorical), Discrete vs Continuous, Structured vs Unstructured, and Primary vs Secondary data. Understanding these classifications helps in choosing appropriate analytical methods.

What is data cleaning, and why is it important?

Data cleaning involves identifying and correcting errors, inconsistencies, and missing values in datasets. It's crucial because poor data quality leads to incorrect analysis and flawed business decisions. Clean data ensures reliable insights and accurate predictions.

Statistics Questions

Explain the difference between population and sample.

A population includes all members of a defined group, while a sample is a subset of the population. Sampling allows us to make inferences about the entire population without analysing every member, making analysis more efficient and cost-effective.

What is correlation, and how is it different from causation?

Correlation measures the strength and direction of a relationship between two variables. Causation implies that one variable directly causes changes in another. The key principle: correlation does not imply causation. Two variables can be correlated due to a third variable or a mere coincidence.

Explain different types of sampling methods.

Common sampling methods include: Simple Random Sampling, Systematic Sampling, Stratified Sampling, and Cluster Sampling. Each method has specific use cases depending on the population characteristics and research objectives.

What is a pivot table, and when would you use it?

A pivot table is a data summarization tool that allows you to reorganize and summarize selected columns and rows of data. It's useful for creating quick summaries, identifying patterns, and

performing cross-tabulation analysis without writing complex formulas.

What are measures of central tendency?

The three main measures are: Mean (average value), Median (middle value when data is sorted), and Mode (most frequently occurring value). Each measure provides different insights and is suitable for different types of data distributions.

What is standard deviation, and why is it important?

Standard deviation measures the spread or dispersion of data points from the mean. It's important because it helps understand data variability, identify outliers, and assess the reliability of the mean as a representative measure.

How would you identify outliers in data?

Outliers can be identified using: Statistical methods (Z-score, IQR method), Visualization techniques (box plots, scatter plots), and Domain knowledge. The $1.5 \times \text{IQR}$ rule is commonly used: values below $Q1 - 1.5 \times \text{IQR}$ or above $Q3 + 1.5 \times \text{IQR}$ are considered outliers.

What is data validation, and why is it necessary?

Data validation ensures data accuracy, completeness, and consistency. It involves checking data against predefined rules, formats, and constraints. It's necessary to maintain data quality, prevent errors from propagating through analysis, and ensure reliable insights.

Why do you want to become a data analyst?

Focus on your passion for solving problems through data, interest in uncovering insights that drive business decisions, and desire to work at the intersection of technology and business. Mention specific aspects of data analysis that excite you, such as pattern recognition or predictive modelling.

What is the difference between Type I and Type II errors?

Type I error (false positive) occurs when we reject a true null hypothesis. Type II error (false negative) happens when we accept a false null hypothesis. The significance level (α) controls Type I error probability, while power ($1-\beta$) relates to Type II error probability.

Explain the Central Limit Theorem and its importance.

The Central Limit Theorem states that the sampling distribution of the mean approaches a normal distribution as sample size increases, regardless of the population distribution. It's crucial because it allows us to make inferences about population parameters using normal distribution properties, even with non-normal populations.

What is the difference between parametric and non-parametric tests?

Parametric tests assume specific distributions (usually normal) and use population parameters. Non-parametric tests make no distributional assumptions and use ranks or medians. Examples: t-test vs. Mann-Whitney U, ANOVA vs. Kruskal-Wallis.

When would you use a chi-square test?

Chi-square tests are used for categorical data to test independence between variables or goodness of fit to expected distributions. Common applications include testing whether two categorical variables are related or if observed frequencies match expected frequencies.

Explain the assumptions of linear regression.

Linear regression assumes linearity (the relationship between variables is linear), independence (observations are independent), homoscedasticity (constant variance of residuals), normality (residuals are normally distributed), and no multicollinearity among predictors.

Mixed Questions

What are the most important Excel functions for data analysis?

Key functions include VLOOKUP/XLOOKUP for data retrieval, SUMIFS/COUNTIFS for conditional aggregation, INDEX/MATCH for flexible lookups, PIVOT functions for summarization, statistical functions (AVERAGE, MEDIAN, STDEV), and text functions (LEFT, RIGHT, MID) for data cleaning.

How do you handle large datasets in Excel?

Techniques include using Excel tables for better performance, applying filters to work with subsets, using array formulas efficiently, minimizing volatile functions, splitting large datasets across multiple sheets, and considering Power Query for data processing.

Explain the difference between VLOOKUP and INDEX/MATCH.

VLOOKUP searches for values in the leftmost column and returns values from columns to the right. INDEX/MATCH is more flexible, allowing lookups in any direction, handling column insertions better, and generally performing faster with large datasets.

How do you create dynamic charts in Excel?

Dynamic charts can be created using named ranges with the OFFSET function, Excel tables (which automatically expand), dynamic array formulas, or combo boxes with data validation to change chart data sources dynamically.

What is Power Query, and how is it useful?

Power Query is Excel's data transformation tool that allows importing, cleaning, and reshaping data from various sources. It's

useful for automating repetitive data preparation tasks, combining data from multiple sources, and maintaining data refresh workflows.

How do you identify and handle duplicates in Excel?

Duplicates can be identified using conditional formatting, the Remove Duplicates feature, or formulas like COUNTIF. Handling strategies include removing exact duplicates, highlighting for manual review, or using advanced criteria to determine which duplicates to keep.

Explain different types of Excel charts and their use cases.

Common chart types: Column charts for comparing categories, Line charts for trends over time, Scatter plots for correlations, Pie charts for parts of a whole, Bar charts for ranking, and Combo charts for multiple metrics with different scales.

Explain the difference between WHERE and HAVING clauses.

WHERE filters rows before grouping occurs and cannot use aggregate functions. HAVING filters groups after GROUP BY and can use aggregate functions. Example: WHERE filters individual sales records; HAVING filters groups like total sales by region.

What are the different types of JOINS, and when do you use them?

INNER JOIN returns matching records from both tables. LEFT JOIN returns all records from the left table and matched records from the right. RIGHT JOIN does the opposite. FULL OUTER JOIN returns all records from both tables. CROSS JOIN creates a Cartesian product.

What is the difference between the RANK () and DENSE_RANK () functions?

RANK () leaves gaps in ranking when there are ties (1, 2, 2, 4), while DENSE_RANK () doesn't leave gaps (1, 2, 2, 3). ROW_NUMBER () assigns unique sequential numbers regardless of ties.

How do you handle NULL values in SQL?

NULL handling functions include: IS NULL/IS NOT NULL for checking, COALESCE for returning the first non-null value, ISNULL/IFNULL for replacing nulls, and NULLIF for creating nulls. NULLs require special consideration in comparisons and calculations.

Explain the difference between UNION and UNION ALL.

UNION combines results from multiple SELECT statements and removes duplicates. UNION ALL combines results but keeps all duplicates. UNION ALL is faster as it doesn't perform deduplication.

How do you optimize SQL query performance?

Optimization techniques: use appropriate indexes, avoid SELECT *, use WHERE clauses to limit rows, use JOINS instead of subqueries when possible, analyse execution plans, avoid functions in WHERE clauses, and use LIMIT for large result sets.

What are Common Table Expressions (CTEs) and their benefits?

CTEs are temporary named result sets defined within an SQL statement using the WITH clause. Benefits include improved readability, recursive operations, multiple references within the same query, and better organization of complex queries compared to subqueries.

Tips for Success in Data Analyst Interviews

Succeeding in data analyst interviews requires more than just technical knowledge. Here are strategic tips specifically for Indian freshers:

- **Technical Preparation:** Focus on fundamentals rather than trying to learn every advanced technique. Master Excel, SQL basics, and one visualization tool thoroughly. Create a portfolio showcasing your analytical projects using publicly available datasets.

- **Communication Skills:** Indian employers highly value analysts who can communicate effectively with diverse stakeholders. Practice explaining technical concepts in simple terms. Prepare examples of how you've presented data insights to different audiences.
- **Industry Awareness:** Stay informed about the Indian data analytics landscape. Understand which industries are hiring (IT, banking, e-commerce, telecommunications), familiarize yourself with major Indian companies using analytics (TCS, Infosys, Flipkart, Paytm), and know about government initiatives like Digital India and their data implications.
- **Cultural Considerations:** Indian work environments often value collaborative problem-solving and respect for hierarchy. Show your ability to work in teams, respect for senior colleagues' experience, and willingness to learn from others. Demonstrate adaptability, as Indian businesses often require analysts to wear multiple hats.
- **Salary Negotiations:** Research current market rates for data analyst positions in your target cities. Entry-level positions in tier-1 cities typically offer ₹3-8 lakhs annually, while tier-2 cities may offer ₹2.5-5 lakhs. Consider the complete compensation package, including benefits, learning opportunities, and career growth prospects.

Excel Question

1. What's the difference between VLOOKUP and HLOOKUP?

VLOOKUP searches for a value in the first column of a table and returns a value in the same row from a specified column. HLOOKUP does the same but searches horizontally in the first row and returns a value from a specified row.

2. How do you apply conditional formatting in Excel?

Select the cells you want to format, go to the Home tab, click Conditional Formatting, choose a rule type (like highlighting cells greater than a value), and set the format you want to apply.

3. How do you remove duplicates from a column in Excel?

Select the data range, go to the Data tab, click Remove Duplicates, choose the columns to check for duplicates, and press OK.

4. What does the IF function do?

The IF function checks a condition and returns one value if true and another if false. Example: =IF(A1>10, "Yes", "No") returns "Yes" if A1 is greater than 10, otherwise "No".

5. What are absolute vs. relative cell references?

Relative references change when copied (e.g., A1 becomes A2). Absolute references stay fixed when copied, marked by \$ signs (e.g., \$A\$1 always refers to cell A1).

6. How do you freeze panes in Excel?

Go to the View tab, click Freeze Panes, and choose to freeze the top row, first column, or a specific section to keep it visible while scrolling.

7. What's the shortcut to select an entire column?

Press Ctrl + Space to select the entire column where the cursor is located.

8. How do you use SUMIF?

SUMIF adds values in a range that meet a specific condition. Example: =SUMIF (A1:A10, ">5") adds all numbers greater than 5 in the range A1 to A10.

9. What are named ranges in Excel?

Named ranges are user-defined names for a cell or range of cells, making formulas easier to read and manage. You can create them via Formulas > Define Name.

10. What is a cell address in Excel?

A cell address is the location of a cell identified by its column letter and row number, like A1 or B5

11. How do you insert or delete rows and columns?

Right-click the row or column header, then select Insert or Delete from the menu.

12. What is the difference between a workbook and a worksheet?

A workbook is the entire Excel file, which contains one or more worksheets (individual sheets or tabs within the file).

13. How do you create a drop-down list using data validation?

Select the cells, go to Data > Data Validation, choose List under Allow, and enter the options separated by commas.

14. What is the purpose of the CONCATENATE function?

It joins two or more text strings into one. For example, =CONCATENATE(A1, " ", B1) combines the contents of A1 and B1 with a space between them.

15. How do you create a pivot table?

Select your data range, go to Insert > PivotTable, choose where to place it, and then drag fields into Rows, Columns, Values, and Filters to summarize data.

16. What is the difference between a line chart and a scatter plot?

A line chart shows trends over time by connecting data points with lines. A scatter plot shows the relationship between two variables using dots.

17. How do you use COUNTIF?

COUNTIF counts the number of cells that meet a condition.

Example: =COUNTIF (A1:A10, ">5") counts how many cells have values greater than 5.

18. What is the order of operations in Excel formulas?

Excel follows PEMDAS: Parentheses, Exponents, Multiplication/Division, Addition/Subtraction.

19. How do you copy formulas without changing references?

Use absolute references (with \$) to fix cells, or use shortcuts like Ctrl + D (fill down) or Ctrl + R (fill right) to copy formulas.

20. How do you handle missing data in Excel?

Use Conditional Formatting to highlight blanks, then fill missing values with averages or other methods using formulas like =IF(ISBLANK(A2), AVERAGE (\$A\$2: \$A\$100), A2).

21. What is the difference between a function and a formula in Excel?

A formula is a user-written calculation (like =A1+B1), while a function is a built-in formula provided by Excel (like SUM () or AVERAGE ()).

22. How do you clear the cell formatting without removing text?

Go to the Home tab, click the Clear button, and select "Clear Formats." This removes formatting but keeps the cell content.

23. How do you enter the current date and time on an Excel sheet?

Use =NOW () for the current date and time. For just the date, press Ctrl +; and for just the time, press Ctrl + Shift +;

24. What is a Dashboard?

A dashboard is a visual summary of important information, often using charts and color-coded indicators to help users quickly review and interpret data.

25. How do you split information in a column into two or more columns?

Use the “Text to Columns” feature under the Data tab. You can split by a delimiter (like a comma or space) or by a set number of characters.

These questions and answers cover key Excel basics that interviewers commonly ask data analyst freshers. Practicing these will help you build confidence and demonstrate your Excel skills effectively.

What Is Microsoft Excel?

Microsoft Excel is a spreadsheet software capable of organizing raw data into a readable format and manipulating it by performing calculations and simple computations.

What Are Cells?

Cells are crosspoints of the rows and columns within a spreadsheet and are referenced by first the column letter and then the row number (i.e. A1). Cells store all information inputted into a spreadsheet, and this information is referred to as “cell content.” Cell content encompasses multiple types of information such as a numeric value, a row or column title, notes, formulas, or functions.

Explain the Spreadsheet and Its Fundamentals.

A spreadsheet in Microsoft Excel is made up of cells organized into rows and columns. Information is inputted into the cells in order to organize, display and manipulate data. For instance, a spreadsheet can contain the names of a list of products, the price per unit, and the number of units sold.

Statistics Questions

1. What are standard deviation and variance?

Variance and standard deviation both measure the dispersion or spread of a dataset. **Variance** is the average of the squared differences from the mean. It gives a sense of how much the values in a dataset differ from the mean. However, because it uses squared differences, the units are squared as well, which can be less intuitive than the standard deviation. **Standard deviation** is the square root of the variance, bringing the units back to the same as the original data. It provides a more interpretable measure of spread. For example, if the variance of a dataset is 25, the standard deviation is $\sqrt{25} = 5$.

2. What is skewness?

Skewness measures the asymmetry of a dataset around its mean, which can be positive, negative, or zero. Data with positive skewness, or right-skewed data, has a longer right tail, meaning the mean is greater than the median. Data with negative skewness, or left-skewed data, has a longer left tail, meaning the mean is less than the median. Zero skewness indicates a symmetric distribution, like a normal distribution, where the mean, median, and mode are equal.

3. What is a histogram?

A [histogram](#) is a graphical representation of the distribution of a dataset. It divides the data into bins (intervals) and shows the frequency (or count) of data points within each bin. Histograms are used to understand the underlying frequency distribution (shape) of a set of continuous data. They help identify patterns such as skewness, modality (number of peaks), and the presence of outliers.

4. What is the difference between descriptive and inferential statistics?

Inferential statistics involves making predictions or inferences about a population based on a random sample of data taken from that population. It uses various methods to estimate population parameters, test hypotheses, and make predictions. While descriptive statistics summarize and describe the features of a dataset, inferential statistics

use the data to make generalizations and draw conclusions about a larger population.

5. What are the different types of sampling methods?

Different sampling methods are used to ensure samples are representative and random. Simple random sampling gives every member of the population an equal chance of being selected. Systematic sampling involves selecting every k-th member of the population, starting from a randomly chosen point. Stratified sampling divides the population into strata or subgroups, with random samples taken from each stratum. Cluster sampling divides the population into clusters, randomly selecting some clusters and sampling all members within them.

6. What is the central limit theorem?

The central limit theorem states that the sampling distribution of the sample mean will approach a normal distribution as the sample size increases, regardless of the population's distribution, provided the samples are independent and identically distributed.

8. What is a probability distribution?

A [probability distribution](#) describes how the values of a random variable are distributed. It provides a function that maps the outcomes of a random variable to their corresponding probabilities. There are two main types of probability distributions. One is the discrete probability distribution for discrete random variables, such as the binomial distribution or the Poisson distribution. The other is the continuous probability distribution for continuous random variables, such as the normal distribution or the exponential distribution.

9. What is the normal distribution?

The normal distribution, also known as the Gaussian distribution, is a continuous probability distribution characterized by its bell-shaped curve, which is symmetric about the mean. With normal distributions,

the mean is, therefore, equal to the median. Also, it's known that about 68% of the data falls within one standard deviation of the mean, 95% within two standard deviations, and 99.7% within three standard deviations. This is known as the 68-95-99.7 Rule.

10. What is a binomial distribution?

The binomial distribution is a discrete probability distribution that models the number of successes in a fixed number of independent Bernoulli trials, each with the same probability of success. It is used when there are exactly two possible outcomes (success and failure) for each trial. For example, it can be used to model the number of heads in a series of coin flips.

11. What is a Poisson distribution?

The Poisson distribution is a discrete probability distribution that models the number of events occurring within a fixed interval of time or space, where the events occur independently and at a constant average rate. It is appropriate to use when you want to model the count of rare events, such as the number of emails received in an hour or the number of earthquakes in a year.

12. What is a p-value?

A p-value is the probability of obtaining a test statistic at least as extreme as the one observed, assuming the null hypothesis is true. It is used in hypothesis testing to determine the significance of the test result. If the p-value is less than or equal to the chosen significance level (α), we reject the null hypothesis. If the p-value is greater than α , we fail to reject the null hypothesis.

13. What are Type I and Type II errors?

Type I errors in hypothesis testing occur when the null hypothesis is true, but we incorrectly reject it, resulting in a false positive. The probability of making a Type I error is the same as the significance level.

Type II errors occur when the null hypothesis is false, but we fail to reject it, leading to a false negative.

15. What is regression?

[Regression analysis](#) is a statistical method used to examine the relationship between one dependent variable and one or more independent variables. It helps in understanding how the dependent variable changes when any one of the independent variables is varied, while the other independent variables are held constant.

18. What is a confidence interval?

A 95% confidence interval means that if we were to take many samples and calculate a confidence interval for each sample, about 95% of these intervals would contain the true population parameter. We could also say that we are 95% confident that the parameter value lies in the estimated interval.

Power BI Questions

1. What is Power BI?

[Power BI](#) is a business analytics tool developed by Microsoft that helps you turn multiple unrelated data sources into valuable and interactive insights. These data may be in the form of an Excel spreadsheet or cloud-based/on-premises hybrid [data warehouses](#).

You can easily connect to all your data sources and share the insights with anyone.

2. Why should we use Power BI?

Because Power BI provides an easy way for anyone, including non-technical people, to connect, change, and visualize their raw business data from many different sources and turn it into valuable data that makes it easy to make smart business decisions.

3. Difference between Power BI and Tableau.

Power BI and Tableau are both strong tools for [business intelligence](#), but they work best in different situations. Power BI fits well for companies already using Microsoft tools like Excel, Office 365, or Azure because it connects easily with them. Tableau is better if you want highly customizable visuals or need to handle really large datasets.

Power BI can manage big datasets using Direct Query or Premium options, while Tableau uses its in-memory engine to crunch data fast. For beginners and general business users, Power BI is easier to pick up. Tableau gives advanced analysts more control when they want to fine-tune visuals and analytics.

5. What is Power BI Desktop?

Power BI Desktop is a free Microsoft tool that brings together data connections, cleaning, modelling, and visualization in one place. You can connect to various data sources, clean and shape the data with Power Query, and create relationships and calculations with Power Pivot.

9. What is DAX?

DAX stands for Data Analysis Expressions. It's a collection of functions, operators, and constants used in formulas to calculate and return values. In other words, it helps you create new info from data you already have.

16. How would you handle missing data in a Power BI report?

Missing data can throw off your reports and insights, so it's important to handle it carefully in Power BI. Here are some common ways to deal with it:

- Use Power Query's data profiling to spot missing values in your dataset
- Replace missing values with zeros, averages, or the value from the previous or next row
- Remove rows if the missing data is small and doesn't affect the results

- Flag missing values with a calculated column so they're easy to see in reports
- Keep a note of your approach and reasoning, as interviewers often look at how you handle data quality decisions

24. What is the CALCULATE function in DAX?

The CALCULATE function evaluates the sum of the Sales table Sales Amount column in a modified filter context. It is also the only function that allows users to modify the filter context of measures or tables.

26. What is row-level security?

Row-level security limits the data a user can view and access, and it relies on filters. Users can define rules and roles in Power BI Desktop and publish them to the Power BI Service to configure row-level security.

27. Why should you apply general formatting to Power BI data?

You should apply general formatting to make it easier for Power BI to categorize and identify data, making it considerably easier to work with.

54. What are KPIs in Power BI?

KPI stands for Key Performance Indicator. Any professional organization has teams and employees who follow the KPI protocols. The organizations set up KPIs for all the employees. These KPIs act as their targets. These KPIs are compared with previous performance to analyse progress.

55. What is a Slicer?

Slicers are an integral part of a business report generated using Power BI. The functionality of a slicer can be considered similar to that of a filter, but, unlike a filter, a Slicer can display a visual representation of all values, and users will be provided with the option to select from the available values in the slicer's drop-down menu.

58. How should you prepare for a Power BI interview?

Effective preparation involves more than memorizing functions: -

- Understand data modelling, DAX formulas, Power Query transformations, and visualization principles
- Practice building end-to-end reports with real datasets to demonstrate practical skills
- Prepare for scenario-based questions, e.g., handling missing data, creating KPIs, or optimizing slow reports
- Review Power BI Service and Power BI Mobile features, since many interviews test knowledge beyond Desktop

2. Main components of Power BI?

- Power BI Desktop – Create reports
- Power BI Service – Share reports online
- Power BI Mobile – View on phones/tablets
- Power BI Gateway – Refresh data from on-premises
- Power BI Report Server – Host reports on-premises

3. Power BI Desktop vs Power BI Service?

Desktop is for building reports (offline).

Service is for publishing, sharing, scheduling, and collaboration (online).

4. What data sources can Power BI connect to?

Excel, SQL Server, SharePoint, Web, APIs, Azure, Oracle, Google Analytics, and many more.

5. What is DAX?

DAX (Data Analysis Expressions) is a formula language used in Power BI to create custom calculations and expressions like measures and calculated columns.

7. Star vs Snowflake schema?

- Star: Facts connect to dimension tables directly.
- Snowflake: Dimension tables are normalized (split into sub-tables).

8. Calculated Column vs Measure?

- Calculated Column: Row-level, stored in the model.
- Measure: Calculated at query time, based on filters.

9. SUM vs SUMX in DAX?

- SUM: Adds up a column directly.
- SUMX: Iterates through rows, applies an expression, and sums the results.

17. Slicer vs Filter?

- Slicer: Visual-level, user-interactive.
- Filter: Can be applied at visual, page, or report level.

18. What is Drill-through in Power BI?

A feature to jump to a detailed report page with filters based on a selected value.

21. What is Power Query?

A data connection and transformation tool using the M language for cleaning and shaping data before loading it.

25. How to handle nulls in Power Query?

Use "Replace Values", "Remove Rows", or custom logic to fill/replace nulls.

26. What is Row-Level Security (RLS)?

A way to restrict data access for users based on roles – implemented via DAX filters.

27. Dashboard vs Report?

Dashboard: Single page, visual summary.

Report: Multi-page, interactive, built in Desktop.

31. How do you implement row-level security (RLS) in Power BI?

Go to Modelling → Manage Roles → Create Role, define DAX filters (e.g., [Region] = "East"), and publish to Power BI Service to assign users.

32. What are KPIs in Power BI?

KPIs (Key Performance Indicators) are visual cues that show progress toward a measurable goal – created using indicator, target, and status fields.

36. What is the use of the LOOKUPVALUE function?

LOOKUPVALUE retrieves a single value from another table based on a search condition, similar to VLOOKUP in Excel.